Whole genome sequence analyses of Bangladeshi isolate of *Anaplasma marginale* and comparison with global strains at genomic level

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Introduction

This study presents the first whole-genome analysis of a Bangladeshi Anaplasma marginale isolate, a pathogen causing anaplasmosis, a tick-borne disease that reduces cattle productivity, causes anemia, weight loss, abortions, and mortality, and imposes major economic losses. The Chattogram strain showed highest similarity to the Florida strain (97.08% ANI) and clustered within a U.S. clade. Pan-genome analysis identified 429 core and 1,630 accessory genes, while structural comparisons showed overall collinearity with regional variations. These findings reveal genetic diversity, evolutionary relationships, and conserved targets, providing key insights for vaccine development and disease management in Bangladesh

Objective

- To compare the genetic diversity and evolutionary relationships of the Chattogram *A.marginale* isolate with global strains using whole-genome analyses.
- To identify conserved and strain-specific genomic features through pan-genome analysis, synteny mapping, and nucleotide identity assessment.

BUSCO Features	Result
Complete BUSCOs	185
Complete and single-copy BUSCOs	183
Complete and duplicated BUSCOs	2
Fragmented BUSCOs	7
Missing BUSCOs	27
Total BUSCO groups searched	219

Table: A.marginale Chittagong strain BUSCO report

Quast features	Results
GC (%)	49.59
Total length	1185749
N50	1185749
L50	1
Misassemblies	10
Local misassemblies	4
Genome fraction (%)	97.959
Largest alignment	442972
Total aligned length	1177366

Final QUAST featuers

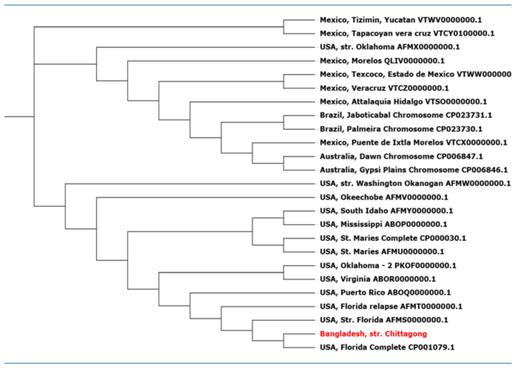
Methodology

01 Blood sample collection from goat, Extraction of DNA upon morphological confirmation from Giemsa staining. PCR amplification using 16s rRNA primer

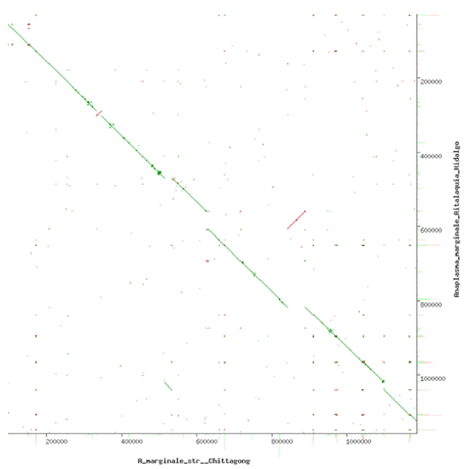
02 Checked quality with FastQC. Removed host reads with Bowtie2. Assembled with SPAdes and RagTag. Evaluated with QUAST and BUSCO

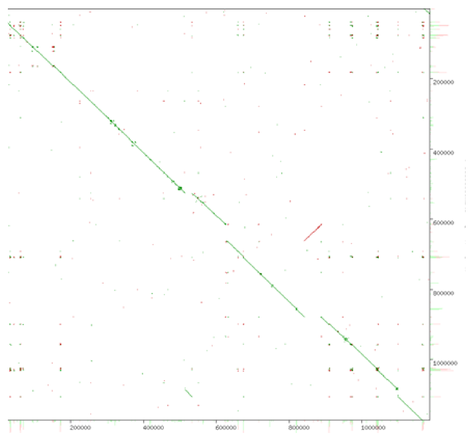
03 Genome Annotation and Submission to NCBI

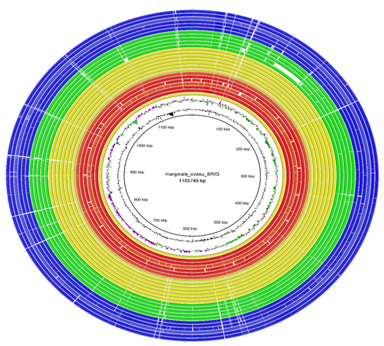
04 Performed ANI, dot plot, BRIG visualization, IQ-TREE phylogeny, Roary pangenome analysis and MSP4 gene comparison



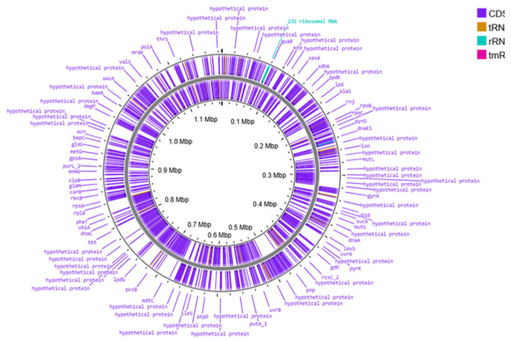
Phylogenetic tree of *A.marginale* based on whole genome alignment







BRIG (Blast Ring Image Generator) Image of Chittagong strain against 24 strains



Prokka annotation plot for A.marginale Chittagong

Results

- Sample & Sequencing
  - Whole-genome sequencing of *A.marginale* from goat blood produced a 1.18 Mb draft genome with 1,010 coding sequences.
  - 98.77% host DNA removed; 189,932 A. marginale-specific reads obtained using a multi-strain reference index.
- Assembly & Annotation
  - Assembly with SPAdes and RAGTAG → near-complete genome (49.59% GC, 185/219 BUSCOs).
  - Prokka annotation identified key genes, including msp4 and Type IV secretion system components.
- Comparative Genomics
  - Average Nucleotide Identity (ANI): 95–97% with 24 global strains.
  - Chittagong strain closely related to Florida, USA strain; divergent from Australian & Mexican strains.
- Phylogenetics & Structural Analysis
  - Evolutionary links to Florida strains confirmed.
  - MSP4 gene highly conserved → reliable phylogenetic marker.
- Pan-Genome Analysis
  - Open pan-genome with extensive strain-specific variability.
- Significance
  - First genome characterization of *A. marginale* in Bangladesh.
  - Findings support vaccine development, surveillance, and control strategies.
- The strain received BioSample ID SAMN50574477 and initial accession number JBQPHU000000000 from NCBI

Conclusion

- First whole-genome sequencing and comparative analysis of goat-derived *A.marginale* isolate from Chittagong, Bangladesh.
- Genetic Similarity-Chittagong strain shows close genetic similarity to Florida, USA strain.Exhibits distinct structural variations from Mexican and Australian strains.
- Pan-Genome Insights-Open genome with extensive accessory genes → indicates regional genomic diversity.
- Possible Introduction-Genetic patterns suggest potential introduction via global cattle trade.
- Significance for Bangladesh-Highlights need to understand local genetic variations for effective control.Supports development of region-specific vaccines, improved diagnostics, and targeted surveillance. Aims to reduce the economic impact of anaplasmosis in Bangladesh.